Procedural Verifiability for Legal AI Outputs: An Audit Framework for Case Recommendation

Abstract

Legal AI case recommendation matters because it can determine which authorities become visible before legal reasons are written. This article argues that such outputs occupy a missing middle category: *normative material screening outputs*. They are more than retrieval results because they structure the materials available for legal argument, but they are not legal sources or decision reasons. Their procedural status should therefore depend on auditability rather than model sophistication, autonomy or aggregate accuracy. The article formalizes procedural qualification through output effect, system role and a six-part audit vector covering source and corpus, query and version, authority hierarchy, ranking and counter-materials, contestability and adoption logging. It implements the model in a provider-agnostic harness and evaluates it across seven layers: stress scenarios, public legal-record snapshots, public-system outputs, raw model outputs, issue-defined positive controls, issue ablations and strict/lenient recoding. The runs show that high authority coverage alone does not create procedural status, source-bound issue packets can qualify as normative screening, and 45 of 47 packet statuses remain stable under strict/lenient recoding. The resulting rule is simple: no reconstructable source chain, no procedural status; no counter-material pathway, no screening qualification; no adoption record, no decision-support reason.

Keywords: legal information retrieval; case recommendation; legal AI auditing; contestability; output evidence packets; normative material screening

1 Introduction

Legal AI systems increasingly decide what law becomes visible. A case recommender, generated research assistant or court-facing authority report may not decide a dispute, but it can set the field of authorities from which reasons are built. That is a procedural act in everything but name. The central risk is not only hallucinated citations or poor retrieval. It is that an output can quietly organize the legal materials available for argument while being treated as a neutral search result.

This article's claim is direct: procedural status should follow auditability, not intelligence. A legal AI output should not be upgraded because the model is autonomous, large, explainable or accurate on average. It should be upgraded only when its source chain, query state, authority hierarchy, counter-material handling, contestability path and adoption record can be reconstructed and challenged. A fluent answer with weak sources remains reference information. A high-scoring ranked list with no counter-material pathway remains professional support. Only a reconstructable and contestable authority-screening chain can become procedural material.

The missing category is *normative material screening*. It captures outputs that filter, rank or summarize legal authorities without becoming legal sources or decision reasons. Many legal AI systems operate exactly there: between search and adjudication, between professional preparation and institutional adoption, and between technical retrieval and legal reasoning. The category also shifts the evaluation question. The issue is no longer whether AI is a legal actor or whether a model is generally capable. The issue is whether a particular output can be given a procedural status above whatever upstream system produced it.

The paper therefore asks three operational questions:

RQ1: How should AI-generated legal outputs be classified according to their procedural effects?

RQ2: What verifiability and contestability requirements are needed for each output class?

RQ3: How can these requirements be operationalized across synthetic, public-output, raw-model, issue-defined and recoding-robustness evaluation settings?

The resulting framework has two dimensions. The output-effect dimension classifies what the output does in the legal process. The system-role dimension classifies how the system enters that process. The two dimensions are then linked to verifiability, contestability and accountability requirements. The model is intentionally output-first: it asks when a legal AI output may move from internal reference to procedural material, and when it must be downgraded or withdrawn.

This article contributes a concept, a model and an artifact. The concept is normative material screening: the intermediate procedural role of AI-generated authority lists, ranked case sets and legal-material summaries. The model separates output effect from system role and links both to a six-part audit vector with status thresholds and failure caps. The artifact is a provider-agnostic harness that implements status allocation, downgrade and withdrawal, threshold sensitivity and coding-uncertainty recoding.

AI-based case recommendation is the core case because it sits at the boundary between legal information retrieval and legal reasoning. Case recommendation systems are central to case-based reasoning and legal information retrieval, both long-standing areas of AI and Law research (Ashley, 1992; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). They are also procedurally sensitive. In common-law precedent search, civil-law court-decision databases, administrative adjudication systems, arbitral award repositories and ODR platforms, ranking and omission can affect the legal materials available for argument. The evaluation below is therefore framed as jurisdiction-neutral: each legal system supplies its own authority hierarchy, but the verifiability problem is shared.

2 Related Work and Gap

2.1 Legal information retrieval and case-based reasoning

Legal information retrieval and case-based reasoning both show why case recommendation cannot be evaluated as generic text ranking. Legal materials do not merely describe an external domain; statutes, precedents, administrative rules and court decisions may constitute or structure the legal domain itself. Legal retrieval must therefore account for hierarchy, temporality, authority, citations and professional use rather than reduce relevance to algorithmic similarity (Turtle, 1995; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). Case-based reasoning reaches the same point from legal reasoning: cases matter through legally relevant similarities, differences, factors, values and precedential relations (Ashley, 1992; Bench-Capon and Sartor, 2003). The procedural risk is that a system may make some authorities salient and others invisible while presenting that ordering as ordinary search.

2.2 Benchmarks and legal retrieval models

Benchmarking work has substantially improved the technical evaluation of legal retrieval. COLIEE defines shared tasks for case-law retrieval, case-law entailment, statute retrieval and statutory entailment, with explicit datasets, runs and leaderboard metrics (Goebel et al., 2024). Neural retrieval systems such as BERT-PLI model paragraph-level interactions between a query case and candidate cases, addressing the length and structure of legal cases more directly than generic ad hoc retrieval models (Shao et al., 2020). Legal language and holding benchmarks such as LexGLUE and CaseHOLD extend evaluation to standardized legal-language understanding and legal holding selection (Chalkidis et al., 2022; Zheng et al., 2021). Recent retrieval-augmented legal benchmarks, including reasoning-focused legal retrieval and CLERC, move closer to professional research tasks by connecting retrieval to downstream legal analysis and citation support (Zheng et al., 2025; Hou et al., 2025). LegalBench further extends the evaluation agenda to a broader set of legal reasoning tasks for large language models (Guha et al., 2023).

These benchmarks are indispensable, but they do not settle the procedural-status question addressed here. Precision, recall, F1, mean reciprocal rank and entailment accuracy evaluate whether a system retrieves or classifies materials correctly under a defined task. They do not determine whether a ranked set may be attached to a court report, relied upon in a professional opinion, or incorporated into a decision reason. A benchmark can show that a system performs well in aggregate while leaving unmeasured whether it records corpus boundaries, distinguishes authority levels, presents counter-authorities, supports party contestation or logs human adoption.

2.3 Generated legal outputs and source verification

Generated legal interfaces make the procedural gap more important. A ranked list exposes a result set for inspection; a generated answer may compress selected materials into a fluent narrative and conceal omissions, conflicts or weak source support. Empirical work on legal hallucinations and commercial legal research tools shows that legal language fluency and retrieval augmentation cannot substitute for source verification (Dahl et al., 2024; Magesh et al., 2025). The audit task is therefore to distinguish source truth, retrieval relevance, authority status and procedural use: an output can be technically useful yet unfit for external procedural status, or partly opaque at the model level yet usable if the legal-material chain is reconstructable and contestable.

2.4 Auditability, explainability and contestability

The framework also draws on algorithmic accountability and AI auditing literature. Work on accountable algorithms emphasizes the need to review data, models and decision pathways rather than accept automated outputs as self-justifying (Kroll et al., 2017). AI auditing scholarship converts this concern into institutional practices for examining systems before and after deployment (Raji et al., 2020; Mökander, 2023). Legal and social-science work on explanation and contestability further shows that explanation matters because affected persons need a meaningful opportunity to understand and challenge consequential decisions (Wachter et al., 2017; Binns et al., 2018).

Existing audit and explainability accounts, however, usually address automated decision systems in general. Due-process and public-sector accountability work shows why affected persons need a practical ability to challenge consequential automated support rather than receive a generic explanation after the fact (Citron and Pasquale, 2014; Veale et al., 2018). That insight has not yet been translated into a domain-specific account of legal AI outputs that structure the visibility and ranking of legal authorities without themselves deciding a case. This article fills that gap by treating case recommendation as normative material screening and by specifying the audit artefacts needed to qualify such outputs for procedural use.

3 Case Retrieval as Normative Material Screening

This article calls such outputs *normative material screening outputs*. The category is not a softer name for retrieval. It identifies an intermediate procedural object: an output that filters, ranks or summarizes legal materials that may later support legal argument or decision-making, without itself creating legal authority or deciding rights and duties. Examples include AI-generated lists of similar cases, recommended statutes, ranked precedent sets, extracted holdings and counter-authority reports.

The distinction also blocks two weak shortcuts. Retrieval metrics such as precision, recall, F1 and mean reciprocal rank are necessary for system development, but they do not say whether a recommendation may be cited in a procedural report, relied upon in a judicial memorandum or incorporated into a decision reason. At the same time, not every legal AI output that influences legal work is a decision-support reason. Many outputs shape the preparation of legal reasoning without becoming reasons for a decision. The middle layer is where most case recommendation systems actually operate.

4 A Two-Dimensional Output Qualification Model

The proposed framework starts from a simple premise: the normative qualification of a legal AI output depends on both what the output does and how the system enters the procedure. These are separate dimensions.

4.1 Output-effect dimension

The output-effect dimension classifies the legal effect claimed for the output:

The classification uses four criteria: whether the output is exposed outside the user's internal workspace, whether a professional relies on it in advice or preparation, whether it filters the set of legal authorities available for argument, and whether an authorized decision-maker adopts it as part of a reasoned outcome. These criteria keep procedural status separate from model quality.

1. **Reference information.** The output supports internal search, summarization or drafting. It has no external procedural effect and is not presented as a basis for another person's legal position.
2. **Professional support output.** The output assists a lawyer, expert, mediator or other professional in forming advice. The professional remains responsible for verification and communication.
3. **Normative material screening output.** The output filters, ranks, summarizes or highlights legal authorities or other normative materials. It may shape the materials available for argument, response or judicial consideration.
4. **Decision-support reason.** The output is adopted, summarized or incorporated by an authorized human or institution as part of the reasons for a legal decision, settlement recommendation, administrative action or arbitral outcome.
5. **No external legal effect.** The output is barred from external legal use because core verifiability, contestability or accountability requirements are absent.

4.2 System-role dimension

The system-role dimension ranges from back-office tools and disclosed assistance tools to auditable procedural tools and unaccountable external disposition tools. The key analytical move is to cross this role with output effect: the same ranked case list may be private reference, court-facing screening, or a decision reason only through authorized adoption.

5 A Verifiability-Based Audit Model

The framework can be expressed as an audit model. Let o be a legal AI recommendation output generated in a particular institutional setting. Its procedural status $PS(o)$ is a function of output effect, system role and an audit vector:

$$PS(o) = g(E(o), R(o), \mathcal{A}(o)).$$

Here $E(o)$ denotes output effect, $R(o)$ denotes system role, and $\mathcal{A}(o)$ denotes a six-part audit vector:

$$\mathcal{A}(o) = (S, Q, H, K, T, L).$$

The six components are source and corpus verifiability (S), query and version traceability (Q), authority-hierarchy representation (H), ranking and counter-material

Table 1: Output status and procedural role

Output class	Typical use	Minimum procedural question	Audit artefact required
Reference information	Internal search, summary, first-pass drafting	Can the user verify the source before relying on it externally?	Source record and user-visible citation or document link.
Professional support output	Lawyer advice, expert preparation, mediator support	Can the professional explain the basis and limitations of the output to the affected person?	Source, query, output and limitation record.
Normative material screening output	Case recommendation, authority ranking, statute or precedent set selection	Can parties or reviewers reconstruct the corpus, version, ranking logic and omitted counter-materials?	Corpus manifest, ranking record, counter-material record and review log.
Decision-support reason	Output adopted in a judgment, administrative decision, arbitral award or ODR recommendation	Can the adoption path, human reasoning and contestation record be audited?	Adoption log, human reasons, contestation record and final responsibility node.
No external legal effect	Unaccountable or unverifiable output affecting rights, duties or procedural opportunities	Is any core requirement missing so that use must be blocked or downgraded?	Failure record stating the missing gate.

verifiability (K), contestability support (T), and adoption logging with audit responsibility (L). Each component is scored on a three-point ordinal scale: 0 means absent, 1 means partially available, and 2 means procedurally reviewable. The model is not a predictive model of legal correctness. It is a status-allocation protocol: it determines the highest procedural status that the output may claim given the audit evidence available.

5.1 Formal definitions and invariants

Let the procedural statuses be ordered as

$$\mathcal{P} = \{p_0, p_1, p_2, p_3, p_4\},$$

where p_0 is no external legal effect, p_1 is reference information, p_2 is professional support output, p_3 is normative material screening output and p_4 is decision-support reason. The order $p_0 \prec p_1 \prec p_2 \prec p_3 \prec p_4$ expresses the maximum procedural status an output may claim. Let $\mathcal{A}(o) \in \{0, 1, 2\}^6$ be the audit vector and let $\Pi = (\tau_N, \tau_D, \Gamma, \Delta)$ be a status policy, where τ_N is the normative-screening threshold, τ_D is the decision-support threshold, Γ is the set of necessary gate predicates and Δ is the failure-flag cap.

Output effect and system role enter the executable protocol by selecting the claimed status, jurisdiction profile and review gate. They are not hidden model features. The operational rule then computes the highest procedurally available status from the audit vector and the relevant adoption gate. For a candidate status p, define a gate predicate $m_p(\mathcal{A}, G_D, \Pi) \in \{0, 1\}$. The preliminary status is the greatest status whose gate is satisfied:

$$\sigma_\Pi(o) = \max_{\preceq}\{p \in \mathcal{P} : m_p(\mathcal{A}(o), G_D(o), \Pi) = 1\}.$$

Let $G_D = 1$ only when the review gate records completed review, authorized adoption, human authorization, jurisdiction assumptions, adoption reasons and contestation record. Failure flags then apply a cap operator $\kappa_F : \mathcal{P} \to \mathcal{P}$:

$$PS_\Pi(o) = \kappa_{F(o)}(\sigma_\Pi(o)).$$

Warning flags leave the status unchanged while recording a defect. Downgrade and suspension flags cap external status at reference information unless the missing artefact is restored. Withdrawal flags cap status at p_0.

The protocol has three formal properties. First, it is monotone only within satisfied gates: if $\mathcal{A}' \geq \mathcal{A}$ componentwise, G_D is unchanged and no failure cap applies, then $g_0(\mathcal{A}') \succeq g_0(\mathcal{A})$; but a missing gate cannot be offset by surplus score elsewhere. Second, caps dominate scores: for any output o, $PS_\Pi(o) \preceq \sigma_\Pi(o)$, and withdrawal flags force $PS_\Pi(o) = p_0$ regardless of upstream recall. Third, upstream metrics are evidence variables rather than status variables. They may inform K or issue-packet construction, but they do not by themselves determine $PS_\Pi(o)$.

5.2 Audit dimensions

Table 2: Audit dimensions for case recommendation outputs

Dimension	Score 0	Score 1	Score 2
S: **Source and corpus**	Sources or corpus cannot be reconstructed.	Individual source links exist, but corpus scope or update status is incomplete.	Source links, corpus manifest, inclusion and exclusion rules, update date and document versions are recorded.
Q: **Query and version**	Query, filters or output version are unavailable.	Some query or version information is retained.	Query, filters, issue labels, source-collection version, workflow or interface version and output identifier are reconstructable.
H: **Authority hierarchy**	Output does not distinguish legal status.	Output marks broad categories such as binding or persuasive.	Output records jurisdiction, court level, legal force, validity, later treatment and relation to governing norms.
K: **Ranking and counter-materials**	Ranking is opaque and contrary materials are not surfaced.	Broad ranking factors or some contrary materials are shown.	Ranking/re-ranking factors, top-k set, omitted-material checks and counter-authority recall are reviewable.
T: **Contestability**	Affected persons cannot inspect or challenge the output.	A professional or internal reviewer can challenge the output.	Parties or reviewers can inspect, challenge, supplement and obtain a recorded response.
L: **Adoption and responsibility**	No adoption path or responsible actor is recorded.	Output and user action are logged.	Output ID, query, result list, adopted and rejected items, reasons, timestamp and responsible reviewer are logged.

This scoring scheme deliberately avoids requiring full model-parameter transparency in every case. A legal procedure may not need source code to decide whether a ranked case list can be contested. It does need enough information to reconstruct the legal-material chain and to test whether a relevant authority was omitted, misranked, outdated or mischaracterized.

5.3 Status thresholds

The audit vector determines the maximum status available to an output. A high aggregate score does not compensate for failure of a necessary gate. The model therefore uses both minimum gates and total-score bands.

Table 3: Status allocation rules

Candidate status	Necessary gates	Additional condition
Reference information	$S \geq 1$ and $Q \geq 1$	Output remains internal or user verifies sources before external reliance.
Professional support output	$S, Q, L \geq 1$	Professional user accepts verification responsibility and can explain limits.
Normative material screening output	$S, Q, H, K, T, L \geq 1$	Total score $\sum \mathcal{A}(o) \geq 9$ and counter-material handling is at least partially reviewable; declared complete-set omissions trigger caps.
Decision-support reason	$S, Q, H, K, T, L \geq 1$, $T = L = 2$ and authorized-adoption gate satisfied	Total score $\sum \mathcal{A}(o) \geq 10$ plus completed review, human authorization, reasons and recorded contestation.
No external legal effect	Any necessary gate for the claimed status is missing	Output is blocked, downgraded or retained only as internal reference.

The mapping function g is therefore explicit rather than discretionary. First, assign the highest status whose gates are satisfied:

$$g_0(\mathcal{A}) = \begin{cases} p_4, & S, Q, H, K, T, L \geq 1,\ \sum \mathcal{A} \geq 10,\ T = L = 2,\ G_D(o) = 1, \\ p_3, & S, Q, H, K, T, L \geq 1 \text{ and } \sum \mathcal{A} \geq 9, \\ p_2, & S, Q, L \geq 1, \\ p_1, & S, Q \geq 1, \\ p_0, & \text{otherwise.} \end{cases}$$

Second, apply failure flags. Withdrawal-level failures, such as unreconstructable sources, materially false generated summaries or irreversible external action without human authorization, set $PS(o)$ to no external legal effect. Suspension or downgrade flags, such as high-authority omission, invalid-authority recommendation, counter-material suppression, missing source-attribution tags, absent jurisdiction assumptions, review-gate failure, ranking drift or contestation failure, cap the output at reference status until the missing artefact is restored. Thus g combines the gate function g_0 with a failure-flag cap. If $K = 0$, the output cannot be treated as normative material screening because the ranking and counter-material treatment cannot be reviewed. If $T < 2$, $L < 2$ or $G_D(o) = 0$, the output cannot be a decision-support reason, even if retrieval accuracy is high. If the system purports to trigger an external disposition without legal authorization, review, contestability or accountability, the output has no external legal effect.

Operationally, an evaluator applies the model in six steps: identify the candidate procedural status claimed for the output; freeze the relevant issue, query, source collection, workflow version and output identifier; score each component of $\mathcal{A}(o)$ using the audit artefacts in Table 2; apply the necessary gates in Table 3; assign the highest available status or record the missing gate; and repeat the audit after material corpus, workflow or interface changes. This makes the framework usable as a deployment checklist, post-use audit and benchmark extension.

The thresholds use non-substitutable gates. The 0–2 scale is ordinal: 0 means absent, 1 means incomplete or internal, and 2 means reviewable by the relevant professional or procedural actor. The default $\tau_N = 9$ permits one limited weakness while still requiring every gate; decision-support status requires $T = L = 2$ because external reasons require both contestation and adoption records. The sensitivity table shows that the default is not a knife-edge choice: thresholds 8–10 produced the same allocation, while threshold 11 moved two normative-screening scenarios to professional support.

Table 4: Sensitivity of stress-test status allocation to threshold policy

τ_N	τ_D	Decision	Normative	Professional	Reference	No effect
8	10	2	3	1	2	2
9	10	2	3	1	2	2
10	11	2	3	1	2	2
11	12	2	1	3	2	2

5.4 Jurisdictional parameterization

The audit dimensions are stable, but the materials satisfying H and K are local. For each setting j, evaluators instantiate authority hierarchy as H_j and counter-material handling as K_j before scoring.

Table 5: Jurisdictional profiles for H_j and K_j

Setting	H_j: authority hierarchy	K_j: counter-materials and limiting materials
Common law	Binding precedent, persuasive precedent, statutes, subsequent treatment, overruled or superseded authorities.	Distinguishing cases, contrary lines, limiting subsequent treatment, minority lines and overruled authorities.
Civil law	Constitutional norms, code provisions, special statutes, regulations, high-court decisions, lower-court decisions and doctrinal commentary.	Competing statutory interpretations, systematic conflicts, teleological limitations, contrary doctrinal views and later amendments.
Admin. adjudication	Statutes, binding regulations, agency rules, official guidance, adjudicative decisions and non-binding policy documents.	Conflicting agency interpretations, withdrawn guidance, higher-level statutory constraints and contrary adjudicative decisions.
Arbitration	Contract text, chosen law, mandatory law, institutional rules, prior awards and trade usage.	Contrary awards, mandatory-law limits, distinguishing clauses, procedural fairness objections and public-policy exceptions.
ODR	Platform rules, consumer statutes, terms of service, regulatory guidance and prior platform decisions.	Consumer-protection overrides, exception policies, appeal decisions and human-review overrides.

5.5 Contestability and audit responsibility

Contestability requires a practical opportunity to inspect selected materials, identify missing or contrary materials, submit alternatives, request human review and receive a recorded response when the output is adopted. Audit responsibility assigns each artefact to the actor that controls it: developers for capability and safety claims, deployers for corpus and interface records, professional users for issue framing and verification, and authorized adopters for decision uptake. The point is status evidence, not a general liability regime.

6 Operationalizing the Framework for Case Recommendation

AI-based case recommendation can be produced by many upstream architectures, but the audit framework does not depend on their internal mechanics. The legally relevant object is the output evidence packet: the source collection, versions, legal materials, output units, source locators, authority labels, counter-material links and adoption records that make the output reconstructable. Failures in these artefacts create different legal consequences. A stale source collection creates source and temporal risk. An ordering rule that privileges surface similarity may miss legally decisive distinctions. A generated summary may misstate holdings or flatten procedural posture. A user interface may display confirming cases while hiding contrary authorities.

The evaluation protocol should therefore separate upstream performance metrics from procedural checks. Precision, recall, F1, MRR and related metrics may help evaluate an upstream retrieval or recommendation component. They do not evaluate whether the output has preserved the legal hierarchy of authorities, displayed counter-materials, distinguished binding and persuasive materials, recorded update status, or produced a contestable adoption trail. These are evaluation requirements for procedural status.

The procedural metrics can be stated directly. *Authority coverage* asks whether known high-authority materials for the issue appear in the output set. *Counter-authority recall* asks whether legally relevant contrary or limiting materials are surfaced. *Hierarchy accuracy* asks whether the output correctly labels binding, precedential, persuasive, overruled or superseded materials. *Version traceability* asks whether the source collection and upstream run state that produced the output can be reconstructed. *Adoption-log completeness* asks whether later human use of the output is observable. These metrics do not replace upstream benchmarks; they determine whether a benchmarked output can be used procedurally.

6.1 Baseline testing

Baseline testing should cover both technical and legal dimensions. A case recommendation system should be tested against known authority sets, especially high-authority or binding materials. Missing a binding, precedential, designated or otherwise high-authority material is different from missing one marginally similar low-level case. The former may trigger withdrawal from procedural use even if aggregate retrieval metrics remain acceptable.

Baseline testing should also record database coverage, update intervals, jurisdictional boundaries, language coverage, case hierarchy, citation treatment and exclusion

Table 6: Decision matrix for legal AI output qualification

Output use	Minimum verifiability	Minimum contestability	Allowed procedural status
Internal legal search	Source and version record; user can inspect retrieved materials	Internal user review	Reference information
Professional advice support	Source, query, output and limitations recorded	Client or supervising professional can request explanation when output materially shaped advice	Professional support output
Court or tribunal case recommendation	Corpus scope, update date, authority hierarchy, ranking factors, counter-authority handling and version record	Parties can inspect, challenge and supplement recommended materials	Normative material screening output
Decision reasoning support	Full adoption path, human confirmation, contrary-material treatment and audit trail	Parties can contest use; decision-maker gives reasons for adoption or rejection	Decision-support reason
Unaccountable external disposition	Missing legal authorization, review, contestability or accountability chain	Not satisfied	No external legal effect

rules. A system trained or indexed on a narrow corpus should not be evaluated as if it were a general legal research system. A system that excludes unpublished decisions, local court decisions or outdated but historically relevant authorities should make that exclusion visible to the user.

6.2 Counter-material presentation

Case recommendation is procedurally dangerous when it confirms one line of reasoning without exposing plausible counter-materials. A useful legal recommendation system should therefore include counter-authority handling as a first-order evaluation requirement. The goal is not to force neutrality in the abstract. The goal is to preserve the adversarial and reason-giving structure of legal reasoning by ensuring that contrary cases, limiting cases, overruled authorities and minority lines are visible when they are legally relevant.

For audit purposes, counter-authority recall can be measured against a manually curated issue set:

$$CAR = \frac{|\text{retrieved counter-authorities} \cap \text{known counter-authorities}|}{|\text{known counter-authorities}|}.$$

The denominator is not the universe of all contrary law. It is the set of contrary or limiting materials identified for a defined issue by the relevant court, professional reviewer or benchmark curator. This makes the measure imperfect but usable.

This requirement is particularly important for generated legal outputs. A ranked list may visibly display multiple authorities. A generated summary may compress them into a single narrative and thereby hide disagreement. Evaluation should test whether the output cites supporting materials accurately, distinguishes holding from dicta, distinguishes binding from persuasive authorities, and signals unresolved conflict instead of hallucinating consensus. Empirical work on legal hallucinations shows why source verification cannot be left to model fluency (Dahl et al., 2024).

6.3 Output evidence packets

The audit layer proposed here sits above any particular search, retrieval, database, generation, agent or manual review architecture. The framework therefore does not inspect upstream implementation details. It requires an *output evidence packet*: a provider-agnostic record that connects the legal propositions or recommended materials in an output to source identifiers, versions, locators and review status.

A minimally reviewable evidence packet should record the issue presented to the system, source collection identifier, source version, output identifier, output units, source identifiers linked to each unit, paragraph or section locators, source-attribution tag, authority labels, counter-material links, human review gate, human adoption record and objection record. This requirement is related to model and dataset documentation work, but the unit of documentation is a procedurally relevant legal output rather than a model card or dataset sheet (Gebru et al., 2021; Mitchell et al., 2019). The upstream system may be a keyword search engine, legal database, generated-output workflow, agent workflow or manual review process. The audit question is the same: can the legal-output chain be reconstructed, source-tagged, reviewed and contested?

Two simple metrics make this requirement operational. *Evidence fidelity* is the proportion of output-source links that support the output unit to which they are attached:

$$EF = \frac{|\text{output-source links that support the unit}|}{|\text{all output-source links}|}.$$

Evidence coverage is the proportion of legally material output units that have at least one reconstructable source locator:

$$EC = \frac{|\text{output units with source-locator support}|}{|\text{all legally material output units}|}.$$

Source-tag coverage is the proportion of output-source links that carry a verification-status tag:

$$STC = \frac{|\text{output-source links with source-attribution tags}|}{|\text{all output-source links}|}.$$

The tag may indicate, for example, whether the source was verified by a connected legal research tool, drawn from public metadata, supplied by the user, treated as settled background, or still requires pinpoint verification. For external normative screening, a tag that merely says "needs verification" is not enough; it records work still to be done rather than procedural source support. Low EF, EC or STC is different from ordinary low upstream recall. It means that the legal output is not procedurally reconstructable or reviewable even if an upstream system found relevant documents. A generated or summarized output with strong upstream performance but weak evidence fidelity, missing locators, untagged sources or no party-facing contestation path should be downgraded or withdrawn from external procedural use.

6.4 Review and adoption gates

Source reconstruction does not by itself authorize legal reliance. The gate record should state whether professional review is required and completed, what jurisdiction assumptions were used, which reliance level applies, whether any irreversible external action has human authorization, and what the human actor did with the output. An accurate draft remains below decision-support status until an authorized actor reviews it, adopts it and records reasons.

The minimum adoption-log schema is: output identifier, time, user role, issue, filters, source-collection version, workflow version, displayed material set, displayed counter-materials, source-attribution tags, review-gate status, adopted item, rejected item, human reason, objection or supplement submitted, response to objection and final responsibility node. A system that cannot export these fields should remain below decision-support status.

7 Downgrade and Withdrawal Conditions

Procedural status is dynamic. Corpus changes, workflow updates, ranking drift, source-attribution gaps, review-gate failures or contestation failures can trigger failure caps after an output or system was initially approved. Table 5 operationalizes those caps: warnings correct isolated defects, downgrades cap status until a missing artefact is restored, suspensions block external use for an issue class, and withdrawals remove external use when the legal-material chain cannot be reconstructed or an irreversible action lacks human authorization.

8 Implementation and Evaluation

To test whether the framework is operational rather than merely classificatory, the rules above were implemented in a lightweight command-line audit harness. The implementation uses local JSON scenario files as the unit of evidence and produces both Markdown and JSON run artefacts. An anonymized companion repository should supply the harness, scenario files, run artefacts and replication commands.

The harness implements the mapping in Section 4 directly. Each scenario declares a claimed status, a base audit vector S, Q, H, K, T, L, a jurisdiction profile, optional authority sets, optional upstream-performance metrics, optional evidence-packet fields, optional review-gate fields and expected outcomes. The runner does not read the expected outcome when computing status. It computes the allowed status, counter-authority recall, authority coverage, invalid-authority rate, packet-level evidence fidelity, evidence coverage, source-tag coverage and downgrade or withdrawal disposition. Separate commands vary status thresholds and re-score the same packets under strict and lenient coding policies.

The complete evaluation matrix contains seven layers: ten protocol stress scenarios, 120 embedded public metadata records, 60 ordered public-system output records,

Table 7: Downgrade severity levels

Severity	Trigger	Procedural consequence
Warning	Isolated error with reconstructable source, query and adoption record.	Correct the output and monitor the issue class.
Downgrade	Missing authority hierarchy, incomplete counter-material display, missing source tags, incomplete review gate or weak adoption logging.	Limit output to reference or professional support status until the missing artefact is restored.
Suspension	Repeated omission of high-authority materials, invalid authority recommendation or unexplained ranking drift in an issue class.	Suspend external procedural use for that domain or data source.
Withdrawal	Source, corpus, ranking or adoption chain cannot be reconstructed, the system generates materially false legal summaries, or an irreversible action lacks human authorization.	Withdraw from external legal use; retain only internal testing or research use.

ten raw model outputs, three issue-defined positive-control packets, twelve issue-level ablations and 94 strict/lenient recoded evaluations over all 47 packets. This is executable audit evaluation rather than a field deployment study. Stress scenarios test the decision protocol. Metadata and public-system outputs test source reconstruction. Raw model outputs test the gap between legal fluency and procedural status. Positive-control packets test the conditions under which normative material screening is justified. Ablations test whether specific defects trigger downgrade or suspension. Recoding tests whether the status allocation is fragile to plausible audit-vector disagreement.

Table 8: Full evaluation matrix

Layer	Units	Rule pass	What it checks
Stress scenarios	10	10/10	Status gates, failure caps, downgrade and withdrawal behavior.
Public metadata	120 records in 6 files	6/6	Source reconstruction across six public legal-record sources.
Public-system outputs	60 records in 6 files	6/6	Real upstream legal-output reconstruction and professional-support caps.
Raw model outputs	10	10/10	High authority coverage without source binding remains procedurally capped.
Positive-control issue packets	3	3/3	Source-bound high-authority and counter-material packets can qualify as normative screening.
Issue ablations	12	12/12	High-authority omissions, counter-material suppression, unverified source tags and missing adoption gates trigger caps.
Annotation robustness	94 recodings	45/47 stable	Strict and lenient recoding policies test status stability across all packets.

Across ten stress-test scenarios, all rule outcomes were reproduced. The mean audit score was 8.30 and mean upstream recall was 0.84. Three scenarios had upstream recall above 0.80 but were procedurally blocked or downgraded. A high-recall output may still be unfit for external legal use when it omits a high-authority material, suppresses a relevant limiting material, recommends invalid authority, lacks output-source evidence or prevents affected persons from contesting the output.

8.1 Public legal-record reconstruction

The harness was also run on public legal-record snapshots. One layer sampled 20 public metadata records from each of six legal systems. A second layer froze the top

Table 9: Scenario-based harness results

Scenario type	Profile	Allowed status	Score	Up. recall	Audit result
Court authority report	Common law	Normative screening	10	0.83	Procedurally usable despite incomplete counter-material recall.
Civil-law statutory interpretation	Civil law	Normative screening	10	0.82	Uses competing interpretations and amendments as counter-materials.
Administrative guidance conflict	Admin.	Reference only	8	0.88	Strong upstream recall but suspended because counter-material and invalid-guidance flags fire.
Grounded output summary	Common law	Normative screening	11	0.90	Output-source mapping and evidence fidelity are complete.
High-coverage uncontestable output	Common law	No external effect	5	0.93	Strong upstream coverage does not cure missing contestability and weak evidence links.
Authorized ODR review	ODR	Decision-support reason	12	0.91	Permitted only because authorization, appeal, contestation and adoption logs are present.

ten records visible in six public legal retrieval or listing streams. These layers use real upstream public legal outputs, but they test reconstruction rather than merits: whether visible records can be frozen, source-tagged, linked to locators and assigned a procedural cap by the audit layer. In both layers, all six scenario files were classified as professional support outputs. The reason is substantive, not conservative: public record visibility establishes source reconstructability, but not issue-specific ranking, counter-material completeness or party-facing contestation.

8.2 Raw model-output pilot

The harness was also run on ten raw legal-authority recommendations generated by a single upstream model configuration, Codex GPT-5.5 with extra-high reasoning. The prompts covered the same three issue families as the positive-control packets: U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, and EU GDPR Article 15 access rights. The model was not allowed to browse or use tools. The point was therefore not to benchmark legal correctness against a commercial research system, but to ask whether fluent model text with strong authority coverage can itself claim normative screening status.

It cannot. All ten raw outputs identified the expected high-authority and limiting materials in the issue families, producing mean upstream recall of 1.00 and counter-authority recall of 1.00. Yet every claimed normative-screening output was downgraded to reference information because the citations were marked as requiring verification rather than bound to official sources, corpus manifests, pinpoint locators and a contestation pathway. The result is the paper's central rule in executable form: authority coverage is not procedural status.

Table 10: Raw model-output pilot

Issue family	Outputs	Mean score	Upstream recall	Allowed status
U.S. agency deference after *Loper Bright*	3	9	1.00	Reference information
English mesothelioma causation after *Fairchild*	3	9	1.00	Reference information
EU GDPR Article 15 access rights	4	9	1.00	Reference information

To test normative material screening itself, the harness includes three manually curated issue packets: U.S. administrative law after *Loper Bright*, English-law mesothelioma causation after *Fairchild*, and EU GDPR Article 15 access-right interpretation. Each packet defines high-authority materials, limiting or counter-materials, invalidated treatments, source tags and output-to-source links at the level of individual output units. These are positive controls, not a benchmark leaderboard: they ask whether a source-bound, issue-defined authority packet can obtain screening status under the protocol. All three qualified as normative material screening, not as decision-support reasons, because no authorized decision-maker adopted them as reasoned outcomes.

Table 11: Issue-defined positive-control packets

Issue packet	Profile	Authority set	Metrics	Status
U.S. agency deference after *Loper Bright*	Common law	H=3; C=2; invalid=1 treatment	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
English mesothelioma causation after *Fairchild*	Common law	H=3; C=2; invalid=1 treatment	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening
EU GDPR Article 15 access rights	Civil law/EU	H=3; C=2; invalid=2 treatments	Auth.=1.00; CAR=1.00; EF=1.00	Normative screening

The ablation suite then removes one procedural condition at a time from those packets. Twelve ablations were generated: high-authority omission, counter-material suppression, unverified source tags and missing authorized-adoption gates for each issue family. Nine were capped at reference information through downgrade or suspension. The remaining three claimed decision-support status but were capped at normative screening because the packet lacked authorized institutional adoption. This is the crucial negative test: a packet that is otherwise strong on source links and authority coverage still loses status when the missing condition is procedural rather than semantic.

The final layer addresses coding uncertainty. All 47 committed packets were rescored under two alternative policies. The strict policy lowers scores when evidence is only internally reviewable, counter-material recall is incomplete, source tags are not procedural, or adoption and contestation records are absent. The lenient policy raises scores only when evidence-packet metrics, authority coverage, counter-authority recall or review gates support the higher score. Status allocation was stable for 45 of 47 packets across base, strict and lenient coding; weighted status agreement was 0.98 for base versus strict and 1.00 for base versus lenient. The two unstable packets were borderline normative-screening stress scenarios that fell to professional support under strict coding, which is the expected direction of error for contested audit evidence.

9 Implications and Limitations

The framework has three implications for legal AI evaluation. First, model performance is not procedural permission. A system may meet technical retrieval benchmarks while failing procedural qualification because the output cannot be reconstructed, source-tagged or contested. Second, human oversight is not magic. Oversight must leave review status, reasons, rejection records and adoption trails; a human signature without verifiable review should not upgrade an output's status. Third, accountability should follow control. Developers, deployers, professional users and institutional adopters control different risks and should carry corresponding explanation duties.

The principal limitation is scale: the current validation is scenario-, artifact- and recoding-based rather than a field deployment study with independent human annotators. It shows how legal-output status can be audited above heterogeneous upstream systems and how robust the status rules are to coding uncertainty, while leaving adoption effects and user behavior for later work.

The next empirical step is therefore clear rather than open-ended. Deployed case recommendation systems and larger legal-output benchmarks should be scored for corpus verifiability, counter-material presentation, ranking traceability, adoption logging and contestation support, and those scores should be compared with retrieval metrics and user behavior. The question for future work is not only whether legal AI systems are accurate, but whether their outputs are legally usable.

10 Conclusion

Status follows auditability, not intelligence. That is the paper's central rule. A case recommendation system may be technically sophisticated, fluent or empirically strong and still have no procedural status if its legal-material chain cannot be reconstructed and challenged. Conversely, an output does not need to be a legal source or autonomous decision-maker to matter procedurally. Once it structures the visibility and ranking of authorities, it occupies the middle category of normative material screening.

The proposed framework makes that category operational. It separates output effect from system role and links both to source, corpus, ranking, counter-material, contestability, adoption and audit requirements. It allows an output to remain internal reference information, become professional support, qualify as normative material screening, or enter a decision reason only through accountable human adoption. It also supplies downgrade and withdrawal rules when the evidentiary and procedural chain fails.

The contribution is therefore not another general claim that AI cannot be a judge or a legal source. It is a stricter evaluation logic for legal AI outputs: no source chain, no status; no counter-material pathway, no screening qualification; no adoption record, no decision reason.

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